

Jay Patel

Java Backend Engineer | Spring Boot | Apache Beam | Large-Scale Data Pipelines | GCP

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Professional Summary

Java Backend Engineer at Tata Consultancy Services specializing in **precision-critical financial data pipelines** and **high-throughput Spring Boot services**. Architected Apache Beam pipelines across 35M+ consumer records with hard numerical accuracy constraints, and engineered **microservices-oriented REST APIs** with fine-grained authentication and authorization controls. Prior background in automotive embedded systems QA adds cross-domain depth to a focused backend engineering profile.

Technical Skills

Languages: Java

Backend & APIs: Spring Boot, Spring WebClient, Project Reactor, REST APIs, Async HTTP, Concurrent Distributed Processing

Data Processing: Apache Beam, Google Dataflow, Batch Processing, Large-Scale ETL

Security: OAuth 2.0, JWT, Spring Security, RBAC, Google KMS, Service Account Impersonation, Authorization

Databases: PostgreSQL, MySQL, JPA / Hibernate, HashMap-based in-memory optimization

Cloud & DevOps: GCP (GCS, Cloud Logging, Dataflow, KMS), AWS, Docker, Maven, Jenkins, CI/CD, GitHub Actions, Git

Certifications: AWS Certified Cloud Practitioner, GitHub Actions, GitHub Foundations

Experience

System Engineer — Java Backend Developer

Tata Consultancy Services

May 2025 – Present

Ahmedabad, India

- **Architected Apache Beam batch pipelines** processing 35M+ consumer records for financial scoring — computing percentile rankings, percentile distributions, and variance with a precision constraint requiring totals to sum exactly to **100%** across all records.
- Eliminated a critical memory bottleneck in the **Risk-Based Pricing pipeline** by replacing bulk data structures with a **HashMap-based credit score aggregation model**, cutting storage from **120 GB to 30 GB** and reducing per-batch runtime from **35 to 20 minutes**.
- Engineered **Spring Boot integration services** driving 1M+ record ML scoring workflows — constructing structured JSON request payloads, dispatching via **async HTTP at 400–500 req/sec**, and processing model responses into structured downstream outputs.
- Leveraged **Apache Beam's distributed worker model on Google Dataflow** for concurrent pipeline execution, enabling parallel processing across large consumer datasets without custom concurrency management.
- Designed and exposed **microservices-oriented REST APIs** for cross-platform enterprise integration, enabling reliable high-volume financial data exchange between internal systems.
- Migrated blocking inter-service communication to a **fully non-blocking reactive model** using Spring WebClient and Project Reactor, measurably improving service throughput and response latency.
- Implemented fine-grained **authorization** using service account impersonation — enforcing read access on source GCS buckets and write access on destination buckets — alongside **OAuth 2.0, Spring Security RBAC, and Google KMS** encryption across all service boundaries.

Assistant System Engineer — Automation & Manual QA

Tata Consultancy Services

Sep 2023 – Apr 2025

Ahmedabad, India

- Authored **automated test scripts in C#** to validate Advanced Driver-Assistance Systems (ADAS) features across IVI GUI simulation platforms, covering functional and regression scenarios for automotive embedded systems.
- Traced and resolved integration defects by analysing Controller-layer message flows in an **MVC-based automotive stack**, collaborating directly with the BOSCH engineering team on cross-layer bug resolution.

Projects

E-Commerce Backend System

Java, Spring Boot, Spring Security, PostgreSQL, JPA/Hibernate

- Designed and built a **complete, independently deployable backend system** covering product catalog, user management, and order lifecycle — structured around a strict **Controller-Service-Repository layered architecture** with clear domain boundaries.
- Implemented a **stateless JWT authentication and authorization system** with role-based access control, centralized exception handling, and a consistent error contract across all API endpoints.
- Modelled the full relational schema using **JPA/Hibernate with PostgreSQL** — defining entity relationships, cascade strategies, and fetch optimizations aligned with real transactional use cases.

Education

Bachelor of Engineering in Computer Engineering

LDRP Institute of Technology and Research

2019 – 2023

Gandhinagar, India